Charitha Raghavaraju

Machine Learning Engineer · Production AI Systems

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ABOUT ME

Machine Learning Engineer with 4 years of experience designing, developing, and deploying production-ready AI solutions using machine learning, deep learning, and Large Language Models (LLMs). Experienced in building scalable AI applications, Retrieval-Augmented Generation (RAG) systems, backend APIs, and cloud-native AI infrastructure on Azure and AWS. Strong software engineering foundation with hands-on experience across audio, speech, and natural language processing, complemented by expertise in Python, cloud technologies, and MLOps practices. Passionate about leveraging AI to solve real-world problems, optimize business processes, and deliver impactful, user-centric solutions.

EDUCATION

M.Sc. Tech — Signal Processing and Machine Learning 2021 – 2024
Tampere University — Tampere, Finland GPA: **4.19 / 5.00**

Thesis: Enhancing Domain-Specific Automated Audio Captioning: Adaptation Techniques and Transfer Learning

B.Tech — Electronics and Telecommunications Engineering 2015 – 2019
National Institute of Technology — Raipur, India GPA: **8.74 / 10.00**

Thesis: Fault Detection using a Hybrid Grey Model, KELM, and Kalman Filter for WSN

PUBLICATIONS

- S. Suzic, I. Martín-Morató, N. Simic, **C. Raghavaraju**, T. Heittola, V. Stanojev, D. Bajovic, "UNS Exterior Spatial Sound Events Dataset for Urban Monitoring," *EUSIPCO 2024 (IEEE)*, Lyon, France, 2024.
- S. Gavel, **C. Raghavaraju**, P. Biswas, A. S. Raghuvanshi, "A data fusion based data aggregation and sensing technique for fault detection in wireless sensor networks" *Computing (Springer)*, vol. 103(11), pp. 2597–2618, 2021.
- P. Biswas, **C. Raghavaraju**, S. Gavel, A. S. Raghuvanshi, "Fault Detection using Hybrid of KF-ELM for Wireless Sensor Networks" *3rd Intl. Conference on Trends in Electronics and Informatics (ICOEI), IEEE*, 2019.

PROJECTS

Document Question-Answering System (RAG) LlamaIndex · LangChain · FAISS
Built a RAG pipeline with dense vector retrieval (FAISS) and LLM reranking for context-aware Q&A. Optimised chunking strategy reduced irrelevant context retrieval and improved answer accuracy on long-form technical documents.

Abstractive Text Summarisation — AWS Production Deployment PEGASUS · Docker · AWS EC2
Fine-tuned Google PEGASUS for domain-specific summarisation, containerised with Docker, pushed to Amazon ECR, and served on EC2, full MLOps cycle from training to scalable cloud inference.

MRI-Based Autism Spectrum Disorder Detection PyTorch · U-Net · ABIDE
U-Net segmentation pipeline on the ABIDE neuroimaging dataset for automated ASD detection, using custom loss functions and data augmentation to handle imbalanced medical imaging data.

TECHNICAL SKILLS

Languages	Python, C++, Java, JavaScript / TypeScript, SQL, Bash
AI / ML / NLP	PyTorch, TensorFlow, Transformers (LLaMA, PEGASUS), RAG, Prompt Engineering, LangChain, LlamaIndex, Hugging Face, Fine-tuning (LoRA / PEFT), NLP, Computer Vision, Audio & Speech ML, Model Evaluation & Testing, vLLM, Ollama
Cloud & MLOps	AWS (EC2, ECR, S3), Azure (Container Apps, Durable Functions, Event Grid, Registry), GCP, Docker, Kubernetes, Terraform, GitHub Actions, MLOps, CI/CD
Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, FastAPI, Streamlit, React, Node.js, Express.js, REST
Databases	PostgreSQL, MongoDB, Vector Databases (Pinecone / ChromaDB / FAISS)
AI Dev Tools	Claude Code, GitHub Copilot, Cursor

EXPERIENCE

AI Engineer Intern Nov 2025 – Dec 2025
Spogen.ai — Tampere, Finland

- Benchmarked **Whisper Large-v3**, **Voxtral**, **Azure speech-to-text (STT)**, and **GPT-4o** on domain-specific Finnish and English audio, applied **AI-coustics enhancement** and **voice activity detection (VAD)** pre-processing, then built an LLM ensemble refinement layer with custom prompts, cutting Finnish **word error rate (WER)** from **35.6%** (Azure STT baseline) to **22.0%**.
- Designed a **domain-adapted GPT-4o transcription pipeline** with terminology-tuned prompts, improving the perfect transcription rate from **44%** to **58%**, and achieving **4.3% WER** with **86.7%** perfect transcriptions on English audio.
- Enabled cost-efficient, on-demand LLM server for the team by deploying a production **vLLM inference server** (NVIDIA T4, OpenAI-compatible API) on **Azure Container Apps** with dynamic **Hugging Face model** loading and fully automated **CI/CD** via GitHub Actions with scheduled container shutdowns.
- Evaluated **ElevenLabs**, **Google text-to-speech (TTS)**, **Chatterbox**, and **Coqui TTS** against audio quality, inference speed, multilingual support, and deployment complexity, delivering a comparative report adopted by the engineering team.

Research Assistant — Audio Research Group Feb 2022 – Dec 2023
Tampere University — Tampere, Finland

- Researched **automated audio captioning (AAC)** and **sound event localisation and detection (SELD)** within the **EU MARVEL project**, spanning model development, synthetic data generation, and a peer-reviewed **IEEE** publication.
- In generic automated audio captioning (AAC), improved the **SPIDEr** score from **0.291** to **0.301** (+3.4%) by applying **layer-wise fine-tuning** on a **patchout fast spectrogram transformer (PaSST) + bidirectional and auto-regressive transformer (BART)** and training domain-specific **byte-pair encoding (BPE)** tokenisers.
- Improved domain-specific captioning accuracy through dataset mixing, parameter-efficient fine-tuning (**PEFT / LoRA**), and systematically comparing audio encoder and caption-decoder configurations to select the best-performing architecture, increasing SPIDEr score from **0.315** to **0.323** on the animal audio dataset (+2.5%) and **0.298** to **0.308** on the vehicle audio dataset (+3.4%).
- Built the **SELD synthetic spatial audio dataset** using **pyroomacoustics** and **FSD50K**, automating onset detection, randomised SNR mixing, and 3D spatial labelling, then trained a SELDnet convolutional recurrent neural network (CRNN) + multi-head self attention (MHSA) model (Multi-ACCDOA output) on it, contributing to the **EUSIPCO 2024** (IEEE) publication.

Backend Software Developer Jul 2019 – Jul 2021
Virtusa Consulting Services (Client: British Telecom (BT)) — Bangalore, India

- Built **OneGWET** (Getafix UI), a BT-internal tool using **Java**, **Spring Boot**, and **Angular 8** that converted Getafix framework XMLs into interactive flowcharts, letting engineers update, delete, and modify business logic through a UI without needing to learn Getafix syntax or touch raw XML. Adopted as the standard tool across BT OpenReach teams.
- Implemented multiple **REST** microservices (flowchart fetch, user profiles, story/release, hotfix, testing, auto Git commit, and rollback) using **Spring Boot** and **MongoDB**, secured with **OAuth 2.0** authentication to control role-based access across the platform.
- Integrated an XML diff checker into the **Angular** frontend, giving engineers a clear side-by-side comparison of XML changes before committing to **GitLab**, reducing review errors and speeding up validation.
- Customised the open-source **Hygieia DevOps** dashboard to surface live performance and security metrics for BT OpenReach projects, giving the client a single consolidated view of project health.

LANGUAGES

English (*Fluent*) · Telugu (*Native*) · Finnish (*Intermediate*) · Norwegian (*Basic*)

REFERENCES

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